

SONICSENTINEL AI - MASTER PROJECT DOCUMENTATION

Multi-Tenant Acoustic Intelligence SaaS Platform | Complete SRS & Execution Blueprint

Theme: AcousticX Intelligence | Category: NextWave AI & ML (Aptech TechWiz 7) | Version: Final 1.0

1. Executive Summary, Problem & SaaS Solution

SonicSentinel AI ek commercial-grade, multi-tenant Acoustic Intelligence SaaS platform hai jo factories, public spaces, buildings, transport hubs, aur homes mein dangerous ya abnormal sounds ko real-time detect karta hai. Purane systems mein insaan ko lagatar audio sunni parti thi jisse response late hota tha, critical sounds miss hoti thin, aur shor mein false alarms aate the. SonicSentinel AI is poore process ko Dual-AI Models (Python ML/DL + Google Teachable Machine) aur Role-Based Multi-Tenant Workflow se automate karta hai.

- UI/UX Standard:** ElevenLabs & Modern AI Studio aesthetic (high-contrast dark studio UI, transport bar, live spectrograms).
- Stripe Billing:** Real Stripe Checkout integration for both Individual (Normal User) and Enterprise (Company) subscriptions.
- Dual-Model Consensus:** Every audio window is independently evaluated by Python ML/DL and Google Teachable Machine (GTM).

2. Multi-Tenant SaaS & Subscription Architecture (B2B + B2C)

Hamara platform 2 tarah ke customers (Subscriptions) ko serve karta hai aur Super Admin poore platform ko control karta hai:

Hierarchy Level	Subscription / Entity	Roles & Operational Flow
1. Super Admin	Platform Owner (Global)	Manages Companies, Subscriptions, AI Model Studio, & Global PI
2. Platform Team	Added by Super Admin	Audio Reviewer, Security & Maintenance staff dedicated to Norm
3. Company (B2B)	Company Subscription	Company Admin adds own Company Security, Maintenance & Audio R
4. Normal User (B2C)	Individual Subscription	Personal monitoring; alerts/reviews handled by Super Admin's P

- 100% Tenant Isolation:** Har Company ka data, sensors, alerts, aur manual reviews doosri company se mukammal isolated rahenge.
- 5 Official SRS Roles:** 1) Normal User, 2) Audio Reviewer, 3) Security Operator, 4) Maintenance Operator, 5) Administrator.

3. Sound Categories, Critical Events & Confusion Pairs

A. 10 Mandatory Core Classes (First Priority)

#	Category Name	Default Severity	Target Role / Routing
1	Gunshot	Critical (Confirmed)	Security Operator (Immediate Tactical Alert)
2	Panic Scream	Critical	Security Operator (Emergency Response)
3	Person Asking for Help	Critical (5 Phrases)	Security Operator (SOS Dispatch)
4	Aggression / Violent Conflict	High / Critical	Security Operator (Conflict Intervention)
5	Glass Breaking	High	Security Operator (Perimeter Breach Alert)
6	Alarm / Siren	High	Security Operator (Attention Alert)
7	Machinery Fault	High	Maintenance Operator (Inspection Recommendation)
8	Vehicle Horn	Low / Medium	Traffic / Environmental Log
9	Animal Sound	Low	Context-Dependent Perimeter Log
10	Background Noise	Informational	Non-Critical Baseline Ambience

- 9 Optional Classes:** Drone, Drilling/Grinder, Fireworks, Vehicle Backfire, Explosion, Door Impact, Graffiti Spray, Normal Machinery, Unknown.
- 7 Confusion Pairs to Distinguish:** Gunshot vs Fireworks/Backfire | Panic Scream vs Shouting | Aggression vs Conversation | Glass Breaking vs Metal Impact | Alarm vs Vehicle Horn | Machinery Fault vs Normal Machinery | Help Request vs Ordinary Speech.

4. Complete UI Portal Modules (Role-Wise Screens)

A. Public Website (Before Login)

- * **1. Landing Page:** ElevenLabs-style AI audio showcase, live interactive demo preview, architecture overview.
- * **2. Pricing & Stripe Subscriptions:** Tab 1: Normal User Plans (B2C) | Tab 2: Company Enterprise Plans (B2B) with Stripe Checkout.
- * **3. Unified Auth Portal:** 'Register as Individual User', 'Register Your Company', and Role-aware Login.

B. Super Admin Portal (Main Platform Owner)

- * **1. Global Command Dashboard:** Total companies, active normal users, Stripe subscriptions, global detections, anomaly warnings.
- * **2. Companies (Tenants) Manager:** View, approve, suspend, or inspect registered B2B companies and their plans.
- * **3. Subscriptions & Stripe Billing:** Manage Normal User & Company plans, payment logs, and data retention policies.
- * **4. Platform Staff Manager:** Create/manage Super Admin's own Audio Reviewer, Security, and Maintenance staff for Normal Users.
- * **5. AI Model Studio & Custom Class Trainer:** Add new sound class, batch-upload clips, auto-extract features, and retrain Python model (v1.0 -> v1.1) + update GTM model link.
- * **6. Dataset & Metadata Registry:** Explore 3,000+ clips manifest (AUD_000001...), 70/15/15 split stats, and 100-Sample Comparison Report.
- * **7. Global Audit Trail & Security Logs:** Track every login, upload, mic session, override, model update, export, and system anomaly.

C. Company Admin Portal (B2B Enterprise Subscriber)

- * **1. Company Executive Dashboard:** Company events, category counts, critical alerts timeline, average confidence, disagreements.
- * **2. 4-Way Audio Studio:** Full access to File Upload, Live Mic, Remote Stream URL, and IoT Sensor/Zone Simulator.
- * **3. Company Staff (Roles) Manager:** Add/manage company's own Security Operators, Maintenance Operators, and Audio Reviewers.
- * **4. Sensors & Zones Manager:** Configure physical/virtual zones (e.g., Gate 1, Boiler Room, Vault) and generate Company API Keys.
- * **5. Custom Alert Rules & Thresholds:** Configure minimum confidence, top-two margin, consecutive window count, and severity actions.
- * **6. Search, Analytics & CSV/Excel Export:** Multi-filter event search, visual charts, forensic PDF/HTML reports, and CSV export.

D. Operational Staff Portals (Company ya Platform ke 3 Specialized Roles)

- * **1. Security Operator Portal:** Live Tactical Threat Radar (Gunshot, Scream, Aggression, Glass, Siren) + Alert Queue (Acknowledge, Dismiss, Escalate).
- * **2. Maintenance Operator Portal:** Machinery Health & Fault Monitor + Inspection Alert Queue + Equipment Audio Analyzer.
- * **3. Audio Forensic Reviewer Portal:** Manual Review Queue + Forensic Workbench (Player, Waveform, Mel Spectrogram, Python vs GTM Top-3) + Override Panel (preserving original AI outputs).

E. Normal User Portal (B2C Individual Subscriber)

- * **1. My Personal Dashboard:** Recent uploads, current detections, personal critical alerts, quality warnings, manual-review status.
- * **2. Personal Audio Studio:** Batch/single upload + Live Mic monitoring + 1-Click SOS Help trigger.
- * **3. My Alerts & Forensic Reports:** View platform reviewer updates and download event reports (with Waveform & Spectrogram).
- * **4. My Profile & Subscription:** Manage User ID profile and Stripe subscription plan.

5. 4-Way Professional Audio Ingestion, Validation & Preprocessing

A. 4-Way Audio Ingestion Hub

Input Mode	Type	Description & Capabilities
1. File & Batch Upload	SRS Mandatory	WAV, MP3, FLAC, OGG, M4A; batch upload; Play, Pause, Replay, Seek, V
2. Live Browser Mic	SRS Mandatory	1-3s sliding windows; browser permission; status: Available/Active/P
3. Stream / URL Fetch	Pro SaaS Feature	Connect live HTTP/RTSP/CCTV audio stream URL or direct cloud recordi
4. IoT API & Simulator	Enterprise SaaS	Company API Keys for edge hardware sensors + Multi-Zone Live Scenari

B. Validation Gate & Audio Quality Grading (4 Grades)

- * **Validation Checks:** Format, file size, duration, sampling rate, channels, corruption check, signal presence check.
- * **Rejection Policy:** Silent, corrupt, unsupported, or too-short files are rejected with clear, human-readable error messages.
- * **Extracted Metadata:** Filename, format, duration, sampling rate, channels, bit depth, file size, upload date/time.
- * **Quality Analysis:** Checks silence, clipping, excessive noise, low signal, encoding issues, missing frames.
- * **4 Quality Grades:** Good | Acceptable | Poor | Unusable (Used directly in Uncertainty and Alert engines).

C. 8-Step Audio Preprocessing Pipeline

Step #	Preprocessing Step	Technical Operation
Step 1	Resampling	Standardize all incoming audio to target sample rate (16,000 Hz).
Step 2	Stereo -> Mono	Convert multi-channel audio into a single unified mono channel.
Step 3	Normalization	Scale peak amplitude into standard dynamic range (-1.0 to 1.0).
Step 4	Silence Trimming	Strip leading and trailing dead air / silence.
Step 5	Noise Reduction	Spectral gating / filtering to reduce stationary background noise.
Step 6	Segmentation	Slice long recordings into fixed windows with exact start/end timestamps.
Step 7	Padding / Truncation	Zero-pad short clips or trim excess samples to exact window length.
Step 8	Format Conversion	Standardize internal buffer to clean 16-bit PCM WAV representation.

6. Dataset Rules, Metadata Manifest & Strict 70/15/15 Split

- * **Minimum Size:** At least 3,000 unique original clips (10 mandatory classes x 300 original clips; our cloud dataset has 7,272 clips).
- * **13-Field Metadata System:** Every clip has: Audio ID (AUD_000001...), filename, category, source, duration, sampling rate, channels, environment, recording device, approximate distance, original/augmented status, dataset split, and SHA-256 hash.
- * **Defined Help Phrases:** 'Help me', 'Somebody help', 'Please help', 'Call for help', 'Emergency' (ethically sourced/voluntary).
- * **Strict 70/15/15 Split:** 70% Training (2,100+ original) | 15% Validation (450+ original) | 15% Testing (450+ original).
- * **Zero Data Leakage:** All segments and augmentations of Recording A MUST stay in the exact same split as Recording A.
- * **Common Dataset Rule:** Both Python and GTM models train and evaluate on the exact same underlying dataset and split.
- * **Augmentation Policy:** Noise, time shift, pitch shift, time stretch, volume, reverb, distance/device sim ONLY in Training; never counted as original clips.

7. Dual-AI Engine, Consensus, Alerts & Manual Review Lifecycle

A. 10 Extracted Acoustic Features & Python 3-Model Bake-off

- * **10 Features:** 1) MFCC, 2) Mel Spectrogram, 3) Chroma, 4) Zero Crossing Rate, 5) RMS Energy, 6) Spectral Centroid, 7) Spectral Bandwidth, 8) Spectral Roll-off, 9) Onset Strength, 10) Tempo.
- * **3+ Models Trained:** Random Forest, XGBoost, and 2D-CNN (with systematic hyperparameter tuning).
- * **Selection Metrics:** Accuracy (85%+), Precision, Recall, F1, Macro F1 (0.80+), Confusion Matrix, Class-wise performance, Critical-event Recall (85%+), and Noise Robustness.
- * **Independent GTM Model:** Google Teachable Machine classifies the same window independently. NEVER pass Python output to GTM.

B. Model Comparison, 4 Consistency Statuses & Uncertainty

- * **Confidence Formula:** Confidence Difference = |Python Top Confidence - GTM Top Confidence| + Top-Two Margin check.
- * **Top-N Display:** UI displays at least the Top 3 predictions from Python and Top 3 from GTM side-by-side.
- * **4 Consistency Statuses:** 1) Acceptable Match | 2) Weak Match | 3) Model Disagreement | 4) Uncertain Result.
- * **5 Uncertainty Triggers:** Low confidence, narrow Top-2 margin, Poor audio quality, Model Disagreement, or Overlapping sounds.
- * **Repeated Confirmation:** Critical alerts require consecutive window confirmation + agreement + acceptable quality.

C. Manual Review Queue, Override Rule & Event Lifecycle

- * **Review Triggers:** Disagreement, low confidence, poor quality, close top-2, overlapping sounds, unknown pattern, or suspected false alarm.
- * **Override Preservation:** Reviewer can override classification, but original Python & GTM outputs are permanently preserved.
- * **Event Lifecycle:** Uploaded -> Classified -> Uncertain -> Alert Generated -> Manual Review -> Reviewed -> Closed.

8. Security, Non-Functional Targets & Step-by-Step Execution Plan

- * **Security & Duplicates:** Controlled audio access, SHA-256 exact duplicate detection, near-duplicate detection, model version linking, full Audit Trail, and Admin Anomaly Alerts.
- * **Performance Targets:** 30s upload processed in < 8s | Live window prediction in < 3s | 20,000+ events scalability | 99% uptime.

Step-by-Step Implementation Roadmap (How We Will Execute Now)

Phase	Deliverable / Action	Key Outcome
Phase 1	Dataset Manifest & 70/15/15 Split	Generate dataset_manifest.csv (AUD_000001...) & data_dictionar
Phase 2	Python 3-Model Training & GTM	Train RF, XGBoost, 2D-CNN on 70/15/15 split + integrate GTM +
Phase 3	SaaS Backend, Stripe & RBAC	Multi-tenant Company/Normal User auth, Stripe Checkout, Custom
Phase 4	4-Way Audio & Consensus Pipeline	Upload, Live Mic, Stream URL, IoT API/Simulator + 8-step prepr
Phase 5	ElevenLabs Studio UI Portals	Build sleek UI for Public, Super Admin, Company Admin, Securit
Phase 6	Testing, Verification & Reports	Run full Pytest suite, verify speed/accuracy targets, finalize